

Ajay I

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Summary

Enthusiastic Computer Science Engineering student skilled in predictive machine learning and building AI agents. Proficient in Python, Java, C and C++, with a proven track record in competitive hackathons and a research publication. Passionate about leveraging advanced ML models and robust architectures to develop scalable, data-driven solutions for real-world challenges.

Education

Sri Ramakrishna Engineering College, Coimbatore B.E. Computer Science and Engineering (Anna University) CGPA: 9.08 / 10 (up to 4th Sem)	2024 – 2028 Coimbatore, IN
Githanjali Public School, Coimbatore (CBSE) HSC (2024) SSLC (2022)	2022 – 2024

Technical Skills

- **Languages:** Java, C, C++, Python
- **AI / ML & Data:** Predictive Machine Learning, NLP (Transformers), Multi-Agent AI, CatBoost, XGBoost, Isolation Forest, Pandas, NumPy, NetworkX, Data Analytics & Visualization, Geospatial Processing
- **Web & Backend:** FastAPI, React.js, Tailwind CSS, JavaScript, REST APIs, GitHub / Slack / Twilio / WhatsApp APIs
- **Databases:** Relational Databases, SQLite, PostgreSQL, Firebase
- **Core Strengths:** Problem Solving, Full-Stack Development, Rapid Prototyping

Projects

Workforce Contribution Monitor

- **Objective:** analyze and bridge the gap between employee activity volume and actual work impact by deriving meaningful contribution metrics.
- Built a full-stack evaluation system integrating data from communication (**Slack**) and execution (**GitHub**) platforms; automated API-based **ingestion pipelines** fetch, process and normalize multi-source data.
- Designed a **custom scoring algorithm** separating high-activity from high-impact contributions, shown on an interactive, responsive dashboard of contribution patterns, trends and performance indicators.
- Focused on modular backend architecture and efficient database handling for scalability, real-time processing and maintainability.

Tech Stack: React, Tailwind CSS, Python, FastAPI, SQLite, GitHub & Slack APIs

AI Customer Feedback Intelligence & Retention System

- **Objective:** analyze customer sentiment, predict churn and automate personalized retention strategies.
- Analyzes multi-channel customer feedback using **Transformer-based NLP** sentiment analysis and behavioral modeling to identify at-risk users.
- Automated pipeline triggering personalized email and WhatsApp follow-ups based on risk levels.
- Integrated a **multi-agent decision framework** recommending refunds, escalations or retention offers, improving decision efficiency.

Tech Stack: Python, NLP (Transformers), FastAPI, React, Twilio API / WhatsApp API, PostgreSQL

Satellite-Based Water Quality Monitoring & Prediction System (Mini Project)

- **Objective:** enable large-scale monitoring and prediction of water quality using satellite data and machine learning.
- Processed satellite imagery and geospatial data to predict key water quality parameters with a **CatBoost Regressor**, using features engineered from spectral and environmental data.
- Built a pipeline for automated ingestion, preprocessing and prediction, with visualization modules for spatial and temporal trend analysis.

Tech Stack: Python, CatBoost, Pandas, NumPy, Geospatial Processing, Data Visualization Tools

ChainWatch – Blockchain AML Intelligence System

- **Objective:** detect suspicious cryptocurrency transactions using machine learning and behavioural analysis.
- Built a real-time system using **XGBoost**, **Isolation Forest**, graph analytics and multi-agent pattern detection to identify high-risk wallet activity and generate explainable risk scores.

Tech Stack: Python, FastAPI, XGBoost, Isolation Forest, NetworkX, Firebase, JavaScript

Publications

[A High-Resolution Data Fusion Framework for Water Body Assessment Using Sentinel-2 and Gradient Boosting](#)
[IEEE Xplore](#) | [DOI](#) | ICCPCT 2026 | IEEE Kerala | 2026

Hackathons & Achievements

- **First Prize** – 24-hr HackTIDE Hackathon, Amrita Vishwa Vidyapeetham, Coimbatore (08.01.2026) | ₹25,000
- **Second Prize** – 12-hr Hackathon, Sri Ramakrishna College of Arts & Science (SRCAS), Coimbatore (17.02.2025) | ₹2,500
- **Third Prize** – 24-hr VeloHack'26, Vel Tech University, Chennai (11.09.2026) | ₹16,000

Certifications

- **Introduction to Artificial Intelligence** – Infosys Springboard
- Cleared **Business English Certificate (BEC)** – Business Preliminary Examination

Declaration: I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief.